SIMATS ENGINEERING

ASSESSMENT TOOL 1: Dashboard Development Task

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO5: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication.

Assessment Tool Weightage: 40%

Total Marks: 20

Code of Conduct

I Naresh M (192424381) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Naresh M

Dashboard Development Task Rubric

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Data Importing, Exploration & Preparation	20	Thoroughly imports, explores, and prepares the dataset; correctly identifies data types, missing values, and relevant variables	Imports and explores data correctly with minor gaps in preparation or identification of issues	Performs basic data exploration and preparation but misses some important aspects	Unable to import or explore the dataset correctly; major preparation issues remain
Python Coding & Pandas/Matplotlib Implementation	25	Effectively applies Pandas and Matplotlib concepts; code is accurate, well-structured, and implements all required analyses	Applies appropriate Pandas and Matplotlib concepts with minor coding or implementation gaps	Limited application of relevant concepts; some required analyses are incomplete	Fails to apply relevant Python concepts; significant coding errors or incomplete implementation
Visualization & Plot Components	25	Selects highly appropriate charts; effectively uses titles, labels, legends, grids, formatting, subplots, and other plot options	Uses appropriate charts and plot components with minor formatting or presentation issues	Uses basic charts but some visualizations or plot components are inappropriate or incomplete	Uses inappropriate/missing visualizations; poor or unclear plot components
Dashboard Design & Analytical	20	Develops a well-organized, coherent dashboard; clearly identifies meaningful	Dashboard is logically organized and identifies relevant	Dashboard has basic organization; identifies some patterns but	Dashboard is poorly organized or incomplete; patterns are incorrectly

Interpretation		trends, relationships, distributions, and patterns with strong justification	patterns with adequate interpretation	interpretation is limited	identified or not interpreted
Output Presentation, Documentation & File Publishing	**10**	Clearly presents results; code is well documented, dashboard is professionally formatted, and output is correctly saved/published	Presents results clearly with minor documentation or output-formatting issues	Basic presentation and documentation; some outputs are unclear or incomplete	Poor presentation; inadequate documentation or failure to produce/save the required dashboard

Total Marks Awarded:

Signature of the Course Faculty

Questions:

5. Retail Customer and Sales Dashboard

A retail store maintains customer and transaction information:

Date, Customer_Age, Gender, Product_Category, Quantity, Sales, Payment_Mode

Develop a **Retail Sales and Customer Analysis Dashboard** using Python.

Task:

1. Import the dataset using Pandas and perform exploratory analysis.
2. Create a **bar chart** showing sales by product category.
3. Create a **count/frequency chart** showing the number of transactions by payment mode.
4. Create a **histogram** showing the distribution of customer ages.
5. Create a **line chart** showing sales evolution over time.
6. Create a **scatter plot** showing the relationship between quantity purchased and sales.
7. Group customers into suitable age categories and compare their sales using a bar chart.
8. Arrange the visualizations into a dashboard using plt.subplots() or plt.subplot().
9. Apply suitable formatting, titles, labels, legends, and figure layout options.
10. Save the final dashboard as an image using plt.savefig().

11. Write a short analytical summary describing the **major customer and sales patterns** discovered.

Aim:

Aim

To develop a Retail Sales and Customer Analysis Dashboard using **Python, Pandas, and Matplotlib** to analyze customer demographics, sales performance, product categories, payment modes, and purchasing patterns.

Dataset:

Date	**Customer_Age**	**Gender**	**Product_Category**	**Quantity**	**Sales**	**Payment_Mode**
2026-01-05	22	Female	Electronics	2	1800	UPI
2026-01-08	35	Male	Clothing	3	2400	Card
2026-01-12	28	Female	Grocery	5	1500	Cash
2026-01-18	42	Male	Electronics	1	1200	Card
2026-01-25	31	Female	Clothing	4	3200	UPI
2026-02-03	25	Male	Grocery	6	1800	Cash
2026-02-10	48	Female	Electronics	2	2500	Card
2026-02-17	37	Male	Clothing	3	2100	UPI
2026-02-22	29	Female	Grocery	4	1400	Cash
2026-03-04	52	Male	Electronics	2	2800	Card
2026-03-11	33	Female	Clothing	5	4000	UPI
2026-03-19	45	Male	Grocery	3	1100	Cash
2026-03-27	26	Female	Electronics	3	2700	UPI
2026-04-05	39	Male	Clothing	2	1600	Card
2026-04-14	55	Female	Grocery	7	2100	Cash

Code:

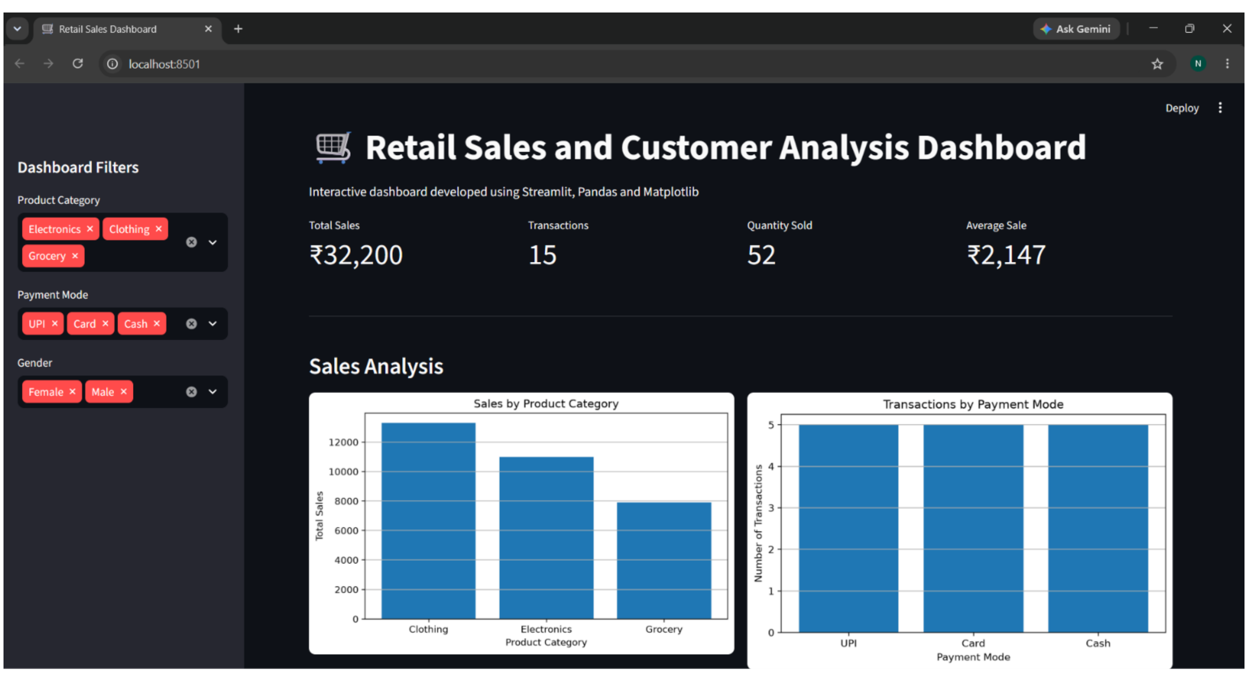
```
import pandas as pd

import matplotlib.pyplot as plt

data = {

    "Date": [

        "2026-01-05", "2026-01-08", "2026-01-12", "2026-01-18",
```

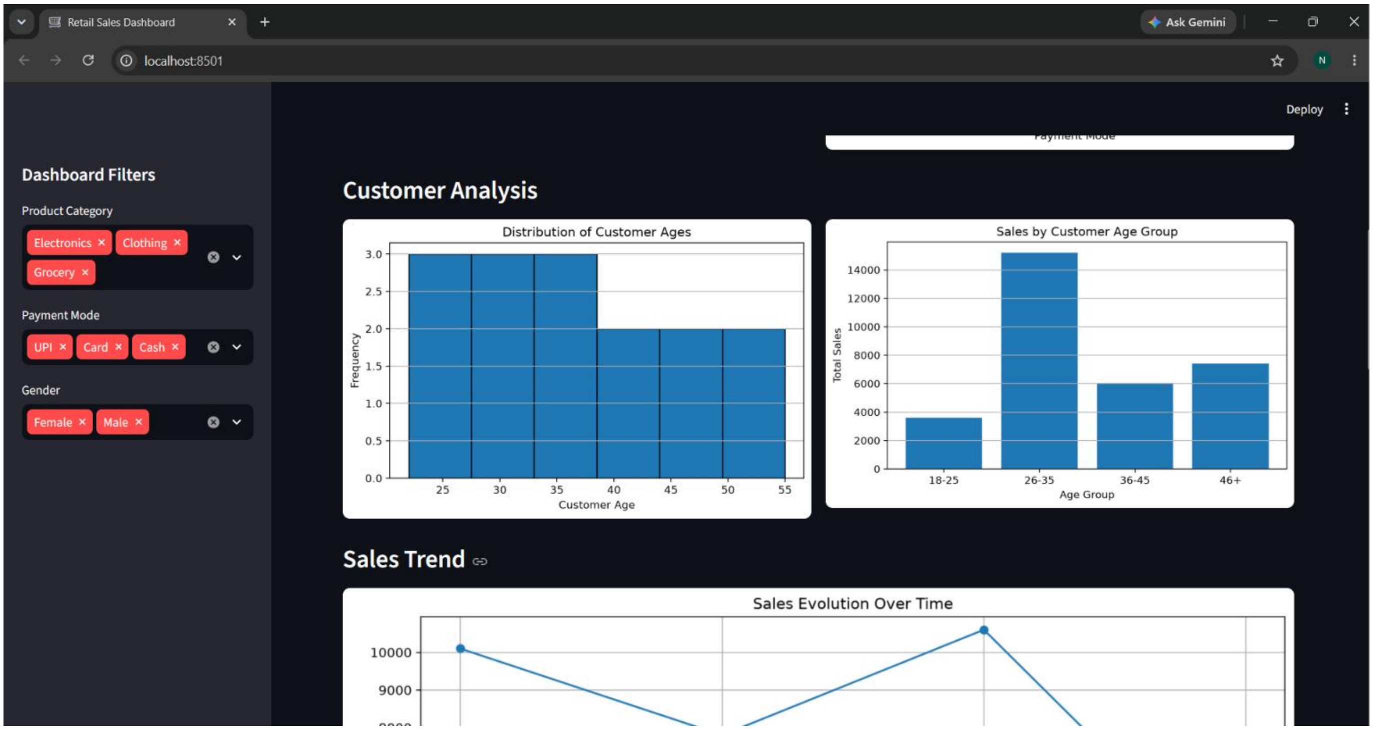
```
        "2026-01-25", "2026-02-03", "2026-02-10", "2026-02-17",
        "2026-02-22", "2026-03-04", "2026-03-11", "2026-03-19",
        "2026-03-27", "2026-04-05", "2026-04-14"
    ],
    "Customer_Age": [22,35,28,42,31,25,48,37,29,52,33,45,26,39,55],
    "Gender": [
        "Female","Male","Female","Male","Female",
        "Male","Female","Male","Female","Male",
        "Female","Male","Female","Male","Female"
    ],
    "Product_Category": [
        "Electronics","Clothing","Grocery","Electronics","Clothing",
        "Grocery","Electronics","Clothing","Grocery","Electronics",
        "Clothing","Grocery","Electronics","Clothing","Grocery"
    ],
    "Quantity": [2,3,5,1,4,6,2,3,4,2,5,3,3,2,7],
    "Sales": [1800,2400,1500,1200,3200,1800,2500,2100,1400,2800,4000,1100,2700,1600,2100],
    "Payment_Mode": [
        "UPI","Card","Cash","Card","UPI",
        "Cash","Card","UPI","Cash","Card",
        "UPI","Cash","UPI","Card","Cash"
    ]
}
df = pd.DataFrame(data)
df["Date"] = pd.to_datetime(df["Date"])
print("First 5 Records:")
print(df.head())
print("\nDataset Information:")
print(df.info())
print("\nStatistical Summary:")
print(df.describe())
```

```
df["Age_Group"] = pd.cut(
    df["Customer_Age"],
    bins=[0, 25, 35, 45, 100],
    labels=["18-25", "26-35", "36-45", "46+"]
)
category_sales = df.groupby("Product_Category", observed=True)["Sales"].sum()
payment_count = df["Payment_Mode"].value_counts()
age_sales = df.groupby("Age_Group", observed=True)["Sales"].sum()
monthly_sales = df.groupby(df["Date"].dt.to_period("M"))["Sales"].sum()
fig, axes = plt.subplots(3, 2, figsize=(14, 16))
axes[0, 0].bar(category_sales.index, category_sales.values)
axes[0, 0].set_title("Sales by Product Category")
axes[0, 0].set_xlabel("Product Category")
axes[0, 0].set_ylabel("Total Sales")
axes[0, 0].grid(axis="y")
axes[0, 1].bar(payment_count.index, payment_count.values)
axes[0, 1].set_title("Transactions by Payment Mode")
axes[0, 1].set_xlabel("Payment Mode")
axes[0, 1].set_ylabel("Number of Transactions")
axes[0, 1].grid(axis="y")
axes[1, 0].hist(df["Customer_Age"], bins=6, edgecolor="black")
axes[1, 0].set_title("Distribution of Customer Ages")
axes[1, 0].set_xlabel("Customer Age")
axes[1, 0].set_ylabel("Frequency")
axes[1, 0].grid(axis="y")
axes[1, 1].plot(
    monthly_sales.index.astype(str),
    monthly_sales.values,
    marker="o",
    label="Sales"
)
```

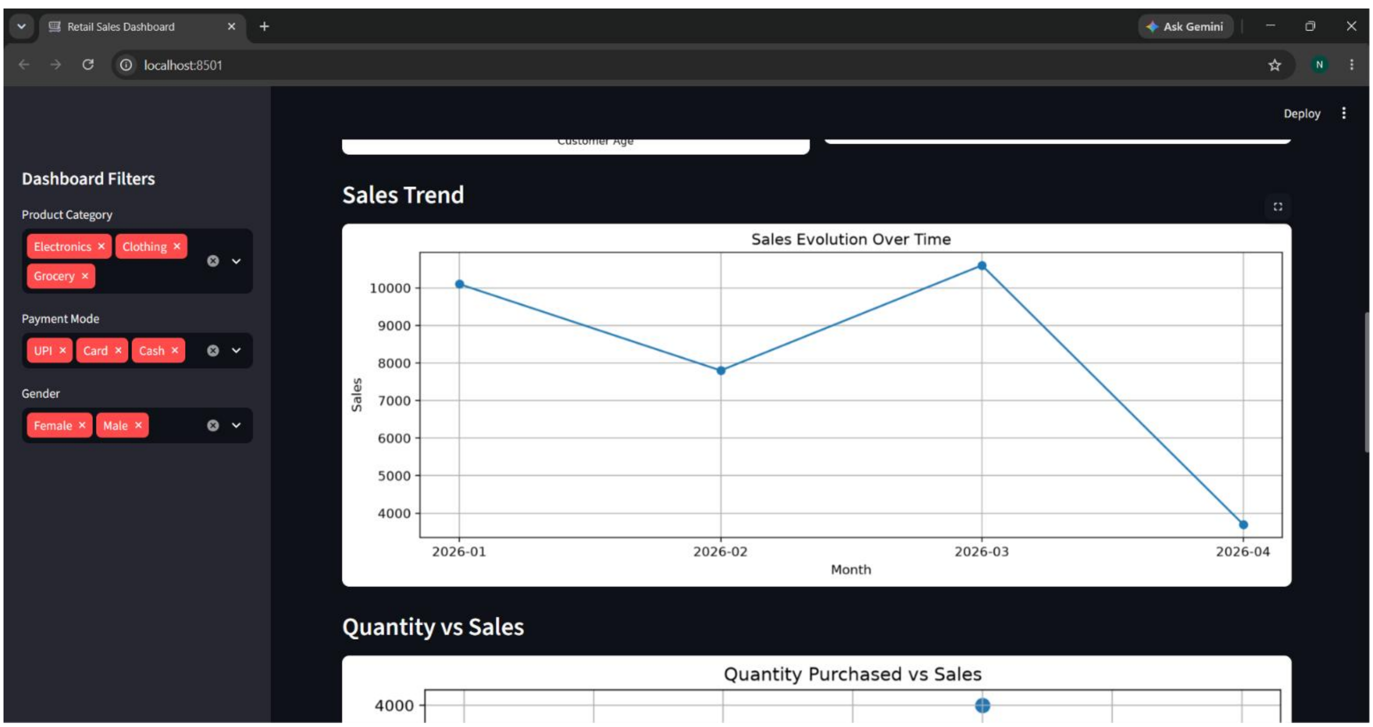
```
axes[1, 1].set_title("Sales Evolution Over Time")
axes[1, 1].set_xlabel("Month")
axes[1, 1].set_ylabel("Sales")
axes[1, 1].legend()
axes[1, 1].grid(True)
axes[2, 0].scatter(df["Quantity"], df["Sales"])
axes[2, 0].set_title("Quantity Purchased vs Sales")
axes[2, 0].set_xlabel("Quantity Purchased")
axes[2, 0].set_ylabel("Sales")
axes[2, 0].grid(True)
axes[2, 1].bar(age_sales.index, age_sales.values)
axes[2, 1].set_title("Sales by Customer Age Group")
axes[2, 1].set_xlabel("Age Group")
axes[2, 1].set_ylabel("Total Sales")
axes[2, 1].grid(axis="y")
plt.suptitle("Retail Sales and Customer Analysis Dashboard", fontsize=18)
plt.tight_layout()
plt.savefig("retail_sales_dashboard.png", dpi=300)
plt.show()
```

Output:





Retail Sales Dashboard
Ask Gemini
localhost:8501
Deploy
Dashboard Filters
Product Category
Electronics
Clothing
Grocery
Payment Mode
UPI
Card
Cash
Gender
Female
Male
Customer Analysis
Distribution of Customer Ages
Frequency
Customer Age
Sales by Customer Age Group
Total Sales
Age Group
18-25
26-35
36-45
46+
Sales Trend
Sales Evolution Over Time



Retail Sales Dashboard
Ask Gemini
localhost:8501
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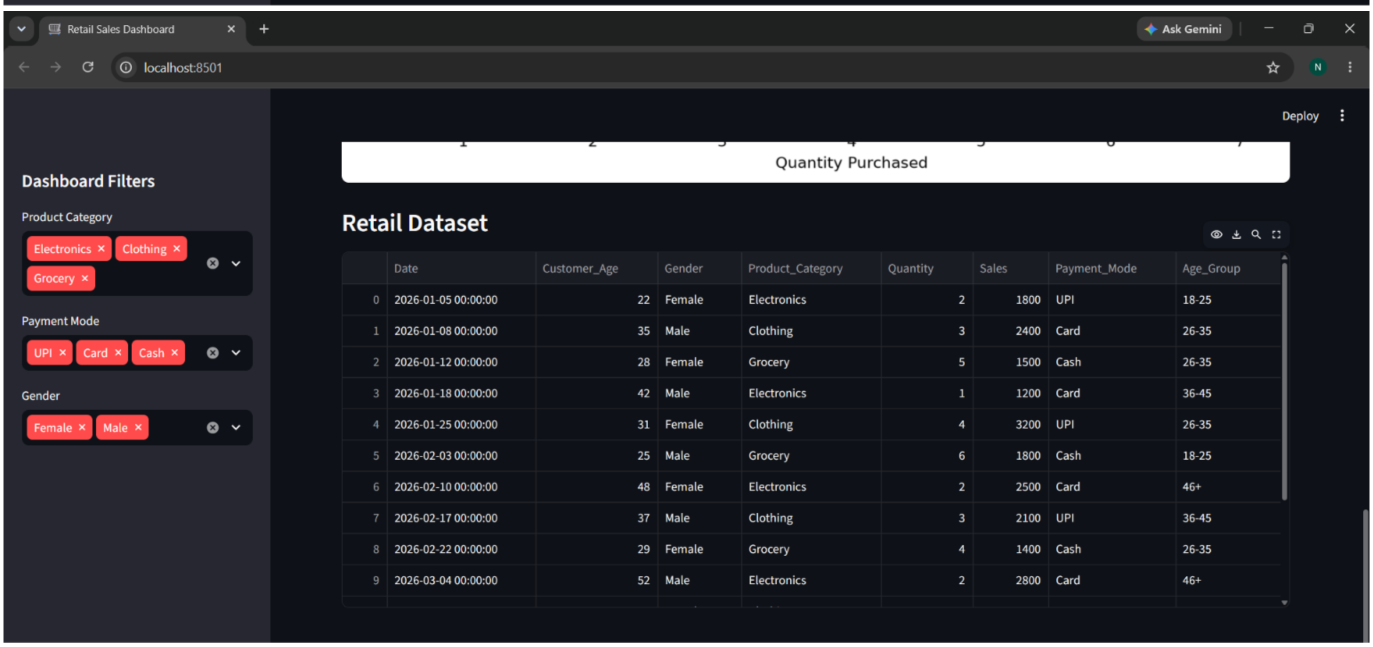
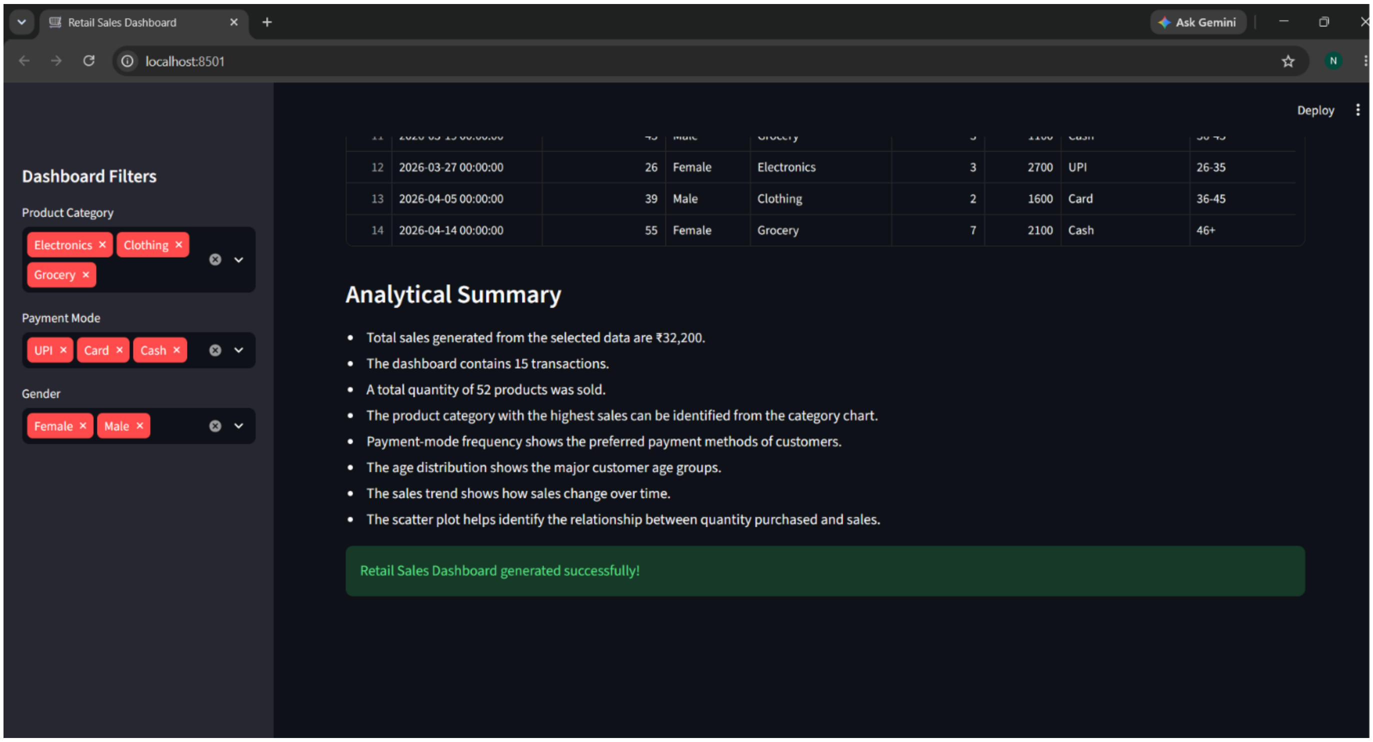
Dashboard Filters
Product Category
Electronics
Clothing
Grocery
Payment Mode
UPI
Card
Cash
Gender
Female
Male
Sales Trend
Sales Evolution Over Time
Sales
Month
2026-01
2026-02
2026-03
2026-04
Quantity vs Sales
Quantity Purchased vs Sales



	Date	Customer_Age	Gender	Product_Category	Quantity	Sales	Payment_Mode	Age_Group
0	2026-01-05 00:00:00	22	Female	Electronics	2	1800	UPI	18-25
1	2026-01-08 00:00:00	35	Male	Clothing	3	2400	Card	26-35
2	2026-01-12 00:00:00	28	Female	Grocery	5	1500	Cash	26-35
3	2026-01-18 00:00:00	42	Male	Electronics	1	1200	Card	36-45
4	2026-01-25 00:00:00	31	Female	Clothing	4	3200	UPI	26-35
5	2026-02-03 00:00:00	25	Male	Grocery	6	1800	Cash	18-25
6	2026-02-10 00:00:00	48	Female	Electronics	2	2500	Card	46+
7	2026-02-17 00:00:00	37	Male	Clothing	3	2100	UPI	36-45
8	2026-02-22 00:00:00	29	Female	Grocery	4	1400	Cash	26-35
9	2026-03-04 00:00:00	52	Male	Electronics	2	2800	Card	46+



Result

Thus, a Retail Sales and Customer Analysis Dashboard was successfully developed using Pandas and Matplotlib. The dashboard provides visual insights into product sales, payment preferences, customer ages, sales trends, purchasing quantity, and customer age groups.